

June 6th, 2024

ISCD - Back Paper - Semester II 23/24

Name (Please Print) _____

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Name _____

Your Signature _____

Instructions:

1. *For writing your answers use both sides of the paper in the answer booklet.*
2. *Additional sheets taken, if any, should be properly attached to the main answer booklet.*
3. **Please write your name on every page of this booklet and every additional sheet taken.**
4. *Maximum time is 3 hours*

Score

Q.No.	Alloted Score	Score
1.	(15 points)	
2.	(15 points)	
3.	(20 points)	
4.	(20 points)	
5.	(15 points)	
6.	(15 points)	
Total	100	

Number of Extra sheets attached to the answer script: _____

1. (15 points) Let $X_1, X_2, \dots, X_n$ be i.i.d. samples from $\text{Exponential}(\lambda)$. Find the maximum likelihood estimate for λ .

-
2. (15 points) Suppose we toss a coin n times with probability p heads. Let $X_i = 1$ if head occurs at the i -th toss and 0 otherwise. Let

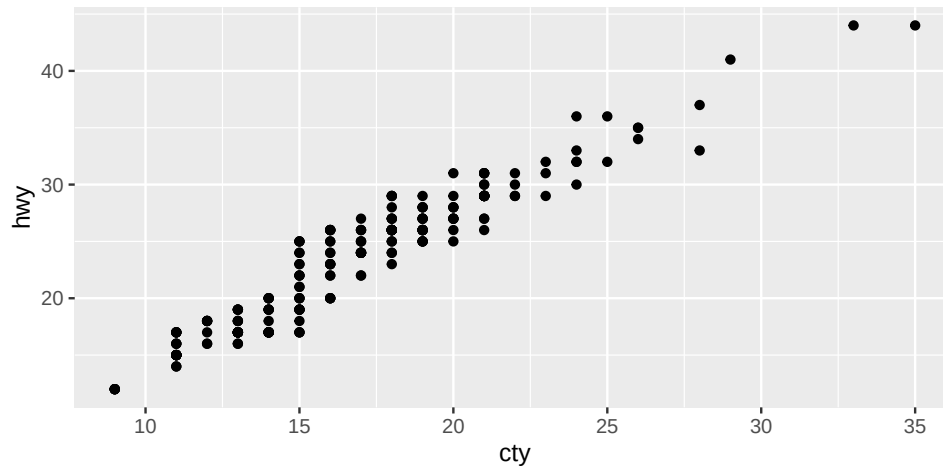
$$\hat{p}_n = \frac{|\{i : X_i = 1\}|}{n}.$$

Find $E[\hat{p}_n]$.

3. Solve the following questions:

- (a) (10 points) What does the plot below tell you about the relationship between city and highway mpg? What does `geom_point()` do?

```
> library(tidyverse)
> ggplot(data = mpg, mapping = aes(x = cty, y = hwy)) +
+   geom_point()
```



- (b) (10 points) Let X be a Normal Random variable with mean μ and standard deviation σ . Compute
- Compute $P(|X - 6| < 3)$ when $\mu = 6$, $\sigma = 6$
 - Compute $P(|X - 1| < 12)$ when $\mu = 1$, $\sigma = 4$
- by using the Normal tables provided.

4. Solve the following questions.

- (a) (10 points) For the R-code below, describe the: null hypothesis, alternate hypothesis, the test statistic, the result of the test (with appropriate justification).

```
> x <- c(0.22, 0.8, 0.5, 0.18, 0.16, -0.67, 1.19, -0.07, -0.13, -0.03)
> y <- c(0.19, 3.00, 1.45, 1.35, 1.95, 1.63, 0.68, 0.51, 1.65, 0.92)
> wilcox.test(x, y)
```

```
Wilcoxon rank sum exact test
```

```
data: x and y
```

```
W = 9, p-value = 0.00105
```

```
alternative hypothesis: true location shift is not equal to 0
```

(b) (10 points) The R-code below considers a sample from a Normal population:

175, 176, 173, 175, 174, 173, 173, 176, 173, 179

Using the outputs of appropriate computation below (with appropriate justification):

(i) describe a confidence interval for the mean of the distribution

(ii) decide on the outcome of the test $H_0 : \mu \geq 175$ vs $H_0 : \mu < 175$

```
> x = c(175, 176, 173, 175, 174, 173, 173, 176, 173, 179)
```

```
> t = sqrt(10)*(mean(x) -175)/(sd(x))
```

```
> pt(t, 9)                                > qt(0.975,9)
```

```
[1] 0.3188211
```

```
[1] 2.262157
```

```
> sqrt(10)
```

```
> mean(x)
```

```
> sd(x)
```

```
[1] 3.162278
```

```
[1] 174.7
```

```
[1] 1.946507
```

5. (15 points) Suppose that U and V are independent $\text{Uniform}(0, 1)$, and interpret them as coordinates of a point in $\mathbb{R}^2$. We wish to estimate the probability that the point (U, V) is inside the unit circle.

- (a) Let $Z = \sqrt{U^2 + V^2}$. Find¹ the $P(Z < 1)$.
- (b) Explain the computation performed in the below **R-code** and (justify) compare the output with answer obtained in part (a).

```
> set.seed(2024)
> I = replicate(10, {
+   u <- runif(10000)
+   v <- runif(10000)
+   sum(u^2+v^2 < 1) / 10000
+ })
> 4*mean(I)

[1] 3.14676
```

¹you may use the fact the integral of constant function 1 over a region in $\mathbb{R}^2$ is the area of the region.

6. (15 points) In an experiment in breeding plants, a geneticist has obtained 219 brown wrinkled seeds, 81 brown roundseeds, 69 white wrinkled seeds and 31 white round seeds. Theory predicts that these types of seeds should be obtained in the ratios 9 : 3 : 3 : 1. Assuming that the null hypothesis is given by the theory, execute a test that can check if there is enough evidence to reject the null hypothesis at 5% level of significance.

Name (Please Print) _____
Statistical Tables for Reference if Needed.

Table 1: in (i, j) -th entry provides $\mathbb{P}(\chi_i^2 \leq j)$, with $\chi_i^2 \sim$ Chi-square with i degrees of freedom.

	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4
2	0.181	0.221	0.259	0.295	0.330	0.362	0.393	0.423	0.451	0.478	0.503
3	0.060	0.081	0.104	0.127	0.151	0.175	0.199	0.223	0.247	0.271	0.294
4	0.018	0.026	0.037	0.049	0.062	0.075	0.090	0.106	0.122	0.139	0.156
5	0.005	0.008	0.012	0.017	0.023	0.030	0.037	0.046	0.055	0.065	0.076
6	0.001	0.002	0.004	0.006	0.008	0.011	0.014	0.018	0.023	0.028	0.034
7	0.000	0.001	0.001	0.002	0.003	0.004	0.005	0.007	0.009	0.012	0.014
8	0.000	0.000	0.000	0.000	0.001	0.001	0.002	0.002	0.003	0.004	0.006
9	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.001	0.001	0.002	0.002

Table 2: in (i, j) -th entry provides $\mathbb{P}(\chi_i^2 \leq j)$, with $\chi_i^2 \sim$ Chi-square with i degrees of freedom.

	2.50	2.52	2.54	2.56	2.58	2.60	2.62	2.64	2.66	2.68	2.70
1	0.886	0.888	0.889	0.890	0.892	0.893	0.894	0.896	0.897	0.898	0.900
2	0.713	0.716	0.719	0.722	0.725	0.727	0.730	0.733	0.736	0.738	0.741
3	0.525	0.528	0.532	0.535	0.539	0.543	0.546	0.549	0.553	0.556	0.560
4	0.355	0.359	0.363	0.366	0.370	0.373	0.377	0.380	0.384	0.387	0.391
5	0.224	0.227	0.230	0.233	0.236	0.239	0.242	0.245	0.248	0.251	0.254
6	0.132	0.134	0.136	0.138	0.141	0.143	0.145	0.148	0.150	0.152	0.155
7	0.073	0.074	0.076	0.077	0.079	0.081	0.082	0.084	0.085	0.087	0.089
8	0.038	0.039	0.040	0.041	0.042	0.043	0.044	0.045	0.046	0.047	0.048
9	0.019	0.020	0.020	0.021	0.021	0.022	0.023	0.023	0.024	0.024	0.025
10	0.009	0.009	0.010	0.010	0.010	0.011	0.011	0.011	0.012	0.012	0.012

Table 3: in (i, j) -th entry provides $\mathbb{P}(\chi_i^2 \leq j)$, with $\chi_i^2 \sim$ Chi-square with i degrees of freedom.

	2.50	2.52	2.54	2.56	2.58	2.60	2.62	2.64	2.66	2.68	2.70
2	0.713	0.716	0.719	0.722	0.725	0.727	0.730	0.733	0.736	0.738	0.741
3	0.525	0.528	0.532	0.535	0.539	0.543	0.546	0.549	0.553	0.556	0.560
4	0.355	0.359	0.363	0.366	0.370	0.373	0.377	0.380	0.384	0.387	0.391
5	0.224	0.227	0.230	0.233	0.236	0.239	0.242	0.245	0.248	0.251	0.254
6	0.132	0.134	0.136	0.138	0.141	0.143	0.145	0.148	0.150	0.152	0.155
7	0.073	0.074	0.076	0.077	0.079	0.081	0.082	0.084	0.085	0.087	0.089
8	0.038	0.039	0.040	0.041	0.042	0.043	0.044	0.045	0.046	0.047	0.048
9	0.019	0.020	0.020	0.021	0.021	0.022	0.023	0.023	0.024	0.024	0.025

Normal Tables
 $\int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp(-\frac{y^2}{2})$

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.10
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	0.5398
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	0.5793
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	0.6179
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	0.6554
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	0.6915
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	0.7257
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	0.7580
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	0.7881
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	0.8159
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	0.8413
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	0.8643
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	0.8849
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	0.9032
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	0.9192
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	0.9332
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	0.9452
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	0.9554
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	0.9641
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	0.9713
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	0.9772
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	0.9821
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	0.9861
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	0.9893
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	0.9918
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	0.9938
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	0.9953
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	0.9965
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	0.9987
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	0.9998